A Constructible Tower Approaching π

Explicit Compass-and-Straightedge Approximants under Transparent Algebraic Acceleration

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Abstract

We exhibit an explicit sequence of compass-and-straightedge constructible real numbers approaching the geometric circle constant $\pi_g = \pi$, although the limit itself is nonconstructible. The base sequence is the Archimedean polygonal quantity

$$P_N = N \sin\Big(\frac{\pi}{N}\Big)$$

restricted to the doubling ladder $N = 4, 8, 16, \ldots$. Every finite term is a constructible length. These terms are combined by the rational extrapolation recurrence

$$T_{i,j} = T_{i,j-1} + \frac{T_{i,j-1} - T_{i-1,j-1}}{4^j - 1},$$

so every tableau entry remains constructible.

The diagonal $T_{i,i}$ has order increasing without bound. A direct interpolation analysis gives

$$\pi - T_{i,i} \sim \frac{\pi^{2i+3}}{(2i+3)!}\, 4^{-(i+1)(i+4)/2}.$$

Consequently, with

$$D(i) = -\log_{10}|T_{i,i} - \pi|,$$

one has

$$D(i) = (\log_{10} 2)i^2 + 2i \log_{10} i + O(i), \qquad \frac{D(i)}{i^2} \longrightarrow \log_{10} 2.$$

The interest is not an arithmetic speed record: the acceleration is classical Richardson–Romberg extrapolation. The point is the explicit combination of field-theoretic constructibility, transparent rational acceleration, and unbounded-order approach to a nonconstructible limit.

1 The Two Facts in Tension

Two facts lie on opposite sides of this construction.

Fact 1: geometric constructibility. For every power of two N, the regular N-gon is constructible by compass and straightedge. Its half-perimeter,

$$P_N = N \sin\Big(\frac{\pi}{N}\Big),$$

is therefore a constructible real number. On the ladder

$$N_i = 4 \cdot 2^i, \qquad i = 0, 1, 2, \ldots,$$

each P_{N_i} is obtained by repeated angle bisection and ordinary chord construction.

Fact 2: the limiting value is not constructible. The polygonal limit is

$$\pi_g := \lim_{i\to\infty} P_{N_i} = \pi.$$

By Lindemann's theorem, π is transcendental and hence is not a constructible real number.

That constructible approximants can approach a nonconstructible limit is not new; Archimedean polygons already do so. The question here is narrower: what happens when the constructible polygonal approximants are accelerated by an explicit rational recurrence that preserves constructibility at every finite stage?

2 The Acceleration Preserves Constructibility

Remark 1 (No circular use of π)**.** *Although the analytic expression $P_N = N\sin(\pi/N)$ contains π, the geometric construction of P_N does not use π as input. It is one half of the perimeter of the regular N-gon inscribed in the unit circle. The displayed trigonometric expression is an analytic description of the constructed length.*

Set

$$T_{i,0} = P_{N_i}, \qquad N_i = 4\cdot 2^i,$$

and for $1 \le j \le i$, define

$$T_{i,j} = T_{i,j-1} + \frac{T_{i,j-1} - T_{i-1,j-1}}{4^j - 1}.$$

The denominators are

$$4^j - 1 = 3, 15, 63, 255, \ldots,$$

and $T_{i,i}$ is the diagonal approximant.

Proposition 1 (Constructibility is preserved)**.** *Every tableau entry $T_{i,j}$, and in particular every diagonal entry $T_{i,i}$, is a constructible real number.*

Proof. Each base term $T_{i,0} = P_{N_i}$ is constructible because N_i is a power of two. The recurrence uses only addition, subtraction, and division by the nonzero rational integer $4^j - 1$. The constructible real numbers form a field, so induction on j proves the claim. □

Thus the diagonal is an explicit tower of constructible numbers. Every finite approximant remains within a finite tower of quadratic extensions, even though the limiting value does not.

3 Interpolation Form of the Diagonal

Introduce

$$x_i = N_i^{-2} = 4^{-(i+2)}$$

and define the entire function

$$F(x) = \begin{cases} \dfrac{\sin(\pi\sqrt{x})}{\sqrt{x}}, & x \neq 0, \\ \pi, & x = 0. \end{cases}$$

Then

$$F(x) = \sum_{m=0}^{\infty} (-1)^m \frac{\pi^{2m+1}}{(2m+1)!} x^m, \qquad F(x_i) = P_{N_i}.$$

The recurrence is Neville–Richardson interpolation at $x = 0$. More precisely, $T_{i,i}$ is the value at 0 of the degree-i polynomial interpolating F at

$$x_0, x_1, \ldots, x_i.$$

It therefore annihilates exactly the monomials

$$x, x^2, \ldots, x^i,$$

which correspond to the polygonal error terms

$$N^{-2}, N^{-4}, \ldots, N^{-2i}.$$

Lemma 1 (Leading diagonal error)**.** *As $i \to \infty$,*

$$\pi - T_{i,i} = \frac{\pi^{2i+3}}{(2i+3)!} \prod_{r=0}^{i} x_r \left(1 + O(i^{-2})\right).$$

Since

$$\prod_{r=0}^{i} x_r = 4^{-(i+1)(i+4)/2},$$

this is equivalently

$$\pi - T_{i,i} \sim \frac{\pi^{2i+3}}{(2i+3)!}\, 4^{-(i+1)(i+4)/2}.$$

Proof. Interpolation is exact for all powers through x^i. For the first omitted monomial x^{i+1}, its interpolant at 0 equals

$$(-1)^i \prod_{r=0}^{i} x_r.$$

The coefficient of x^{i+1} in F is

$$(-1)^{i+1} \frac{\pi^{2i+3}}{(2i+3)!},$$

so the leading contribution to $\pi - T_{i,i}$ is the positive quantity displayed above.

For the remaining Taylor terms, the complete homogeneous symmetric polynomials in $x_0, \ldots, x_i$ are bounded by powers of

$$\sum_{r=0}^{i} x_r < \sum_{r=0}^{\infty} 4^{-(r+2)} = \frac{1}{12}.$$

Successive Taylor coefficients have ratio

$$\frac{\pi^2}{(2m+2)(2m+3)} = O(i^{-2})$$

when $m \geq i + 1$. Hence the entire tail is $O(i^{-2})$ relative to the leading omitted term. □

Definition 1 (Geometric circle constant)**.** *The accelerated constructible tower defines*

$$\pi_g := \lim_{i \to \infty} T_{i,i}.$$

The preceding lemma shows directly that

$$\pi_g = \pi.$$

The notation π_g names the geometric route to the classical circle constant; it does not denote a numerically different constant.

4 Digit Growth

Let

$$D(i) = -\log_{10}|T_{i,i} - \pi|.$$

The leading-error formula gives

$$D(i) = \frac{(i+1)(i+4)}{2}\log_{10} 4 + \log_{10}\big((2i+3)!\big) - (2i+3)\log_{10}\pi + o(1).$$

Using Stirling's formula,

$$D(i) = (\log_{10} 2)i^2 + 2i\log_{10} i + O(i).$$

Therefore,

$$\boxed{\frac{D(i)}{i^2} \longrightarrow \log_{10} 2 \approx 0.3010299957}.$$

The quadratic term comes from the geometric product of the interpolation nodes. The positive $2i\log_{10} i$ correction comes from the factorial decay of the sine-series coefficient. At accessible levels this correction is still substantial, so $D(i)/i^2$ remains above its limiting value.

Level i	$N_i = 4 \cdot 2^i$	Correct digits $D(i)$	$D(i)/i^2$
5	128	19.59	0.78355
10	4096	57.34	0.57337
15	131072	112.05	0.49798
20	4194304	183.12	0.45781
25	134217728	270.26	0.43241
30	4294967296	373.26	0.41474

Table 1: Diagonal of the constructible tower, computed with 600-digit arithmetic. The ratio decreases toward $\log_{10} 2$, not toward $2\log_{10} 2$.

5 What the Construction Means

The claim

There exists an explicit sequence of compass-and-straightedge constructible real numbers generated from the Archimedean doubling ladder by the rational recurrence

$$T_{i,j} = T_{i,j-1} + \frac{T_{i,j-1} - T_{i-1,j-1}}{4^j - 1}.$$

Every finite $T_{i,i}$ is constructible, while

$$\lim_{i\to\infty} T_{i,i} = \pi_g = \pi.$$

The diagonal has unbounded increasing order, and its correct-digit count obeys

$$D(i) = (\log_{10} 2)i^2 + 2i\log_{10} i + O(i).$$

What is not claimed

The construction does not make π constructible, cross the constructible/transcendental boundary, outperform modular-form algorithms in arithmetic complexity, or introduce a new extrapolation method. Richardson–Romberg acceleration of an even-power sequence is classical. The contribution is the explicit constructible provenance of every finite entry and the transparent approach to a boundary that the limit cannot cross.

6 Conclusion

The tower is geometric at its base, algebraic in its acceleration, and analytic only in the verification of its limit and rate. Every finite approximant is constructible; the recurrence uses only rational operations; and the limiting value is the geometric circle constant

$$\pi_g = \pi.$$

The boundary between constructible approximants and the transcendental limit is therefore honored exactly. The tower approaches that boundary with increasing order but never reaches it at any finite stage.

References

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